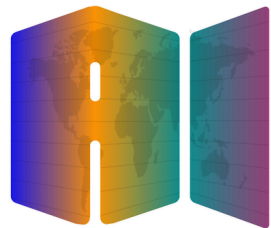




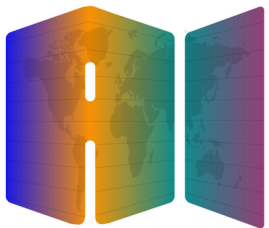
The GATES

AI Ecosystem Concept



The Concept

- A hub where users and contributors can explore functionalities of **GATES MCP server tools** as well as the capabilities of **GATES AI Agents**
- Play around with **Agent Chat Interfaces** and **MCP Inspectors**
- Browse **Quickstart guides** for integration
- Developers can **contribute** tools and agents through MCP and A2A, respectively
- Tools and agents served in the /mcp/servers and /agents endpoints, respectively, are **automatically discoverable** in the hub



A PLACE TO BUILD WHAT'S NEXT

Geospatial intelligence, made discoverable.

Explore a growing ecosystem of specialist tools and AI agents designed to turn location data into clear, useful action.

EXPLORE THE ECOSYSTEM

Where would you like to start?

CURATED COLLECTION 

01



PROTOCOL-READY UTILITIES

MCP Servers

Plug dependable geospatial capabilities directly into your AI workflows.

Browse servers

02



SPECIALIST COLLABORATORS

AI Agents

Meet domain-aware agents that reason through your most complex spatial questions.

Meet the agents

Geospatial capabilities, ready to connect.

Connect your AI workflow to specialist Philippine data and analysis services through one dependable protocol.

 Search servers, topics, or capabilities

FILTER All categories



26 SERVERS



STREAMABLE HTTP

EARTH OBSERVATION

CopPhil MCP Server

Discover and analyse Copernicus satellite scenes for Philippine land, coast, and atmosphere monitoring.

search_sentinel_scenes



STREAMABLE HTTP

ENVIRONMENTAL DATA

OneLab MCP Server

Query accredited laboratory observations for water, soil, air, and environmental samples.

find_lab_observations



STREAMABLE HTTP

WEATHER & CLIMATE

PAGASA TC Bulletin MCP Server

Access tropical cyclone bulletins, forecast tracks, wind signals, and rainfall advisories.

get_active_cyclone_bulletin



STREAMABLE HTTP

RESEARCH & INNOVATION

DOST Projects MCP Server

Explore DOST-funded research, innovation projects, implementation sites, and outputs.

search_dost_projects



STREAMABLE HTTP

DISASTER READINESS

HandaPH MCP Server

Find preparedness resources, incident information, and local disaster-response guidance.

get_preparedness_resources



STREAMABLE HTTP

RISK INTELLIGENCE

GeoRiskPH MCP Server

Screen Philippine locations for multi-hazard exposure using authoritative risk layers.

screen_locations_risk



← Back to catalogue



ENVIRONMENTAL DATA

OneLab MCP Server

Operational

CONNECTION

https://gates.dost.gov.ph/mcp/servers/onelab

Copy endpoint

TRANSPORT

Streamable HTTP

Protocol version 2025-03-26

LIVE SERVER INTERFACE

Connect server

Explore its capabilities.

Query accredited laboratory observations for water, soil, air, and environmental samples.

Tools 3 Resources 2 Prompts 1

AVAILABLE TOOLS

3



find_lab_observations

Retrieve verified laboratory results by location, sample type, and date.



INPUT SCHEMA

{ area_of_interest, date_range }

Run tool



search_onelab_catalog

Search the OneLab catalogue using a location, keyword, or date.



get_data_provenance

Return source metadata, update frequency, and applicable use guidance for a dataset.



GATES / PYTHON QUICKSTART



Connect an MCP Server

Use the official MCP Python SDK with Streamable HTTP.

INSTALL `pip install mcp`

Copy

PYTHON

Copy code

```
import asyncio

from mcp import ClientSession
from mcp.client.streamable_http import streamable_http_client

SERVER_URL = "https://gates.dost.gov.ph/mcp/servers/onelab"

async def main():
    async with streamable_http_client(SERVER_URL) as (read, write, _):
        async with ClientSession(read, write) as session:
            await session.initialize()
            tools = await session.list_tools()
            print([tool.name for tool in tools.tools])

asyncio.run(main())
```

[Read the official documentation](#) ↗

Meet your next spatial collaborator.

Purpose-built collaborators that help make sense of the Philippines through location, data, and expert context.

13 specialised agents

 Search agents or specialties



SPATIAL ANALYSIS

GATES Analytics Agent

• spatial trends • area comparisons



GIS OPERATIONS

GATES GIS Agent

• GIS workflows • map design



ENVIRONMENTAL INTELLIGENCE

OneLab Agent

• water quality • lab results



DISASTER RISK

Hazard Lens Agent

• hazard screening • risk summaries



WEATHER & CLIMATE

Climate Signal Agent

• rainfall context • climate trends



COASTAL & MARINE

Coastal Watch Agent

• coastal exposure • fisheries context



FOOD SYSTEMS

AgriScope Agent



CITIES & MOBILITY

Urban Pulse Agent



EARTH OBSERVATION

Land Cover Agent





+ Explore agents

YOUR AGENT

- GATES Analytics
- GATES GIS
- OneLab
- Hazard Lens
- Climate Signal
- Coastal Watch
- AgriScope
- Urban Pulse
- Land Cover
- Watershed
- Infrastructure Navigator
- Community Access
- PlaceSense

MCP Servers

← Back to GATES

OneLab Agent Environmental Intelligence

Integrate

Switch agent



GATES AI AGENT

How can OneLab help today?

I'm the OneLab Agent. I can help interpret environmental laboratory data and connect water, soil, air, and sampling results to their geographic context.

What can you tell me about water quality?



Help me explore lab results.



Create a brief for a location in the Philippines.



Message your GATES agent...



GATES Agents may make mistakes. Validate information before using it in decisions.



Connect an AI Agent

Discover its Agent Card, then send it a message with the A2A Python SDK.

INSTALL `pip install a2a-sdk`

Copy

PYTHON

Copy code

```
import asyncio

from a2a.client import ClientConfig, create_client
from a2a.helpers import new_text_message
from a2a.types import Role, SendMessageRequest

AGENT_URL = "https://gates.dost.gov.ph/a2a/agents/onelab"

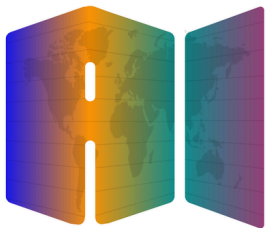
async def main():
    client = await create_client(
        agent=AGENT_URL, client_config=ClientConfig(streaming=False)
    )
    request = SendMessageRequest(
        message=new_text_message("Hello from my app", role=Role.ROLE_USER)
    )
    async for event in client.send_message(request):
        print(event)
    await client.close()

asyncio.run(main())
```

[Read the official documentation](#) 

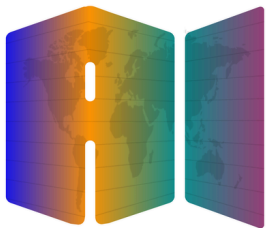
Why Build the AI Ecosystem

- Transform GATES geospatial data, analytics, and visualizations into **discoverable and reusable AI-powered capabilities**
- Enable users to explore complex GIS tools through intuitive MCP Inspectors and AI agent interfaces
- Allow developers and partner agencies to contribute new MCP tools and specialized AI agents
- Create an interoperable environment where agents, tools, and data can work together to support better decision-making



Workshop Rationale

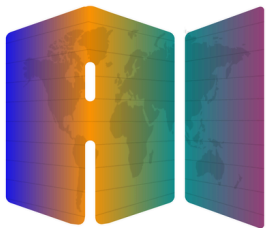
- Provide participants with a practical understanding of **Agentic AI**, **Model Context Protocol (MCP)**, and **Agent-to-Agent (A2A) communication protocol**
- Show users and future contributors what happens **under the hood** when AI agents access tools, data, and other agents
- Build the technical foundation needed to develop, inspect, integrate, and evaluate Agentic Workflows that make use of MCP tools and A2A communication
- Encourage a shared approach to interoperability, security, responsible AI, and system design



From Workshop to Ecosystem

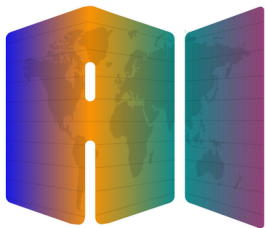
This workshop is the first step towards building a **collaborative** GATES AI community.

- Participants become informed users, developers, and potential contributors
- Workshop outputs and feedback help shape the GATES AI Ecosystem
- Reusable tools and agents can support decision-making at the executive, agency, community, and individual levels
- Over time, the ecosystem grows into a trusted hub for AI-powered geospatial analytics and visualization



From Workshop to Ecosystem

Learn the architectures. Build the capabilities. Grow the ecosystem



Our Goal for Today

Serve and interact with an A2A Agent that has access to MCP tools

But first, we need to build the workflows

